

Lab Sheet 3 (CST-108)

TCP Client-server

1. Write a C / C++ program that does the following:
 - (a) Gets a string from the user
 - (b) Generates a 16 bytes long pseudo random number using Openssl RAND_bytes function
 - (c) Concatenates this with the string provided by the user and uses Openssl functions to generate a SHA256 hash.
 - (d) Print string provided by the user, pseudo random number, and output of SHA256.
2. For this exercise modify TCP server and client programs written in previous lab. Client may request for two different type of services: (i) register at server (ii) request for a file. Server maintains a database of login names and hash of passwords (hash of password || 16-byte salt). Whenever a user registers, it adds an entry for the new user in its database. When client needs a file, it has to first login with correct password. Server verifies the user. Now client sends the name of the file and server sends back the file to user.

[Note: You may use a file (instead of a database) for storing login names, salt and hash separated by “:”.]

3. For this exercise modify TCP server written in previous exercise so that instead of creating a new process it creates a new thread whenever the request (register or login) from a client arrives.

[Note: Use pthread library]